

WEEK 3 POWER BI PROJECT REPORT

Stock Market Analysis Dashboard

Data Analytics using Microsoft Power BI

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1. Introduction

The Week 3 project was developed using Microsoft Power BI to analyze historical stock market data. The dataset contains Date, Open, High, Low, Close, Adjusted Close, and Volume fields. The main purpose of the project was to prepare the data, create useful measures, and build an interactive Stock Market Analysis Dashboard.

Project Objectives

- Prepare and organize the stock market dataset in Power BI.
- Create a separate date table for time-based analysis.
- Create calculated measures for High Price, Low Price, Close Price, Trading Volume, Daily Return, and YTD Close.
- Build cards and charts to display important stock indicators.
- Add a date slicer for interactive filtering.
- Present the results in a clear and professional dashboard.

2. Data and Date Table Preparation

A dedicated DimDate table was created using the minimum and maximum dates from the stock data. This table supports date-based analysis and provides a structured calendar for the dashboard.

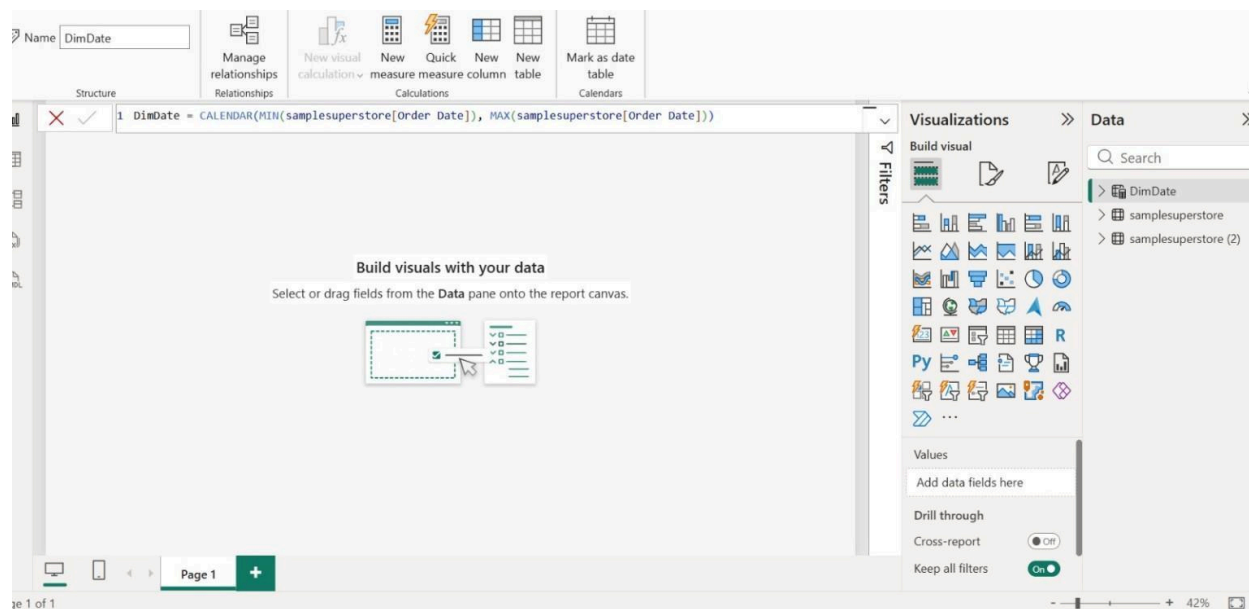


Figure 1: Creation of the DimDate table using the stock data date range.

The DimDate table was created with a calendar range based on the available stock dates. Additional date-related fields were then prepared for analysis.

3. Creating Month Name and Date Fields

A Month Name calculated column was created from the DimDate Date field using the FORMAT function. This helps display months in a readable format and supports time-based reporting.

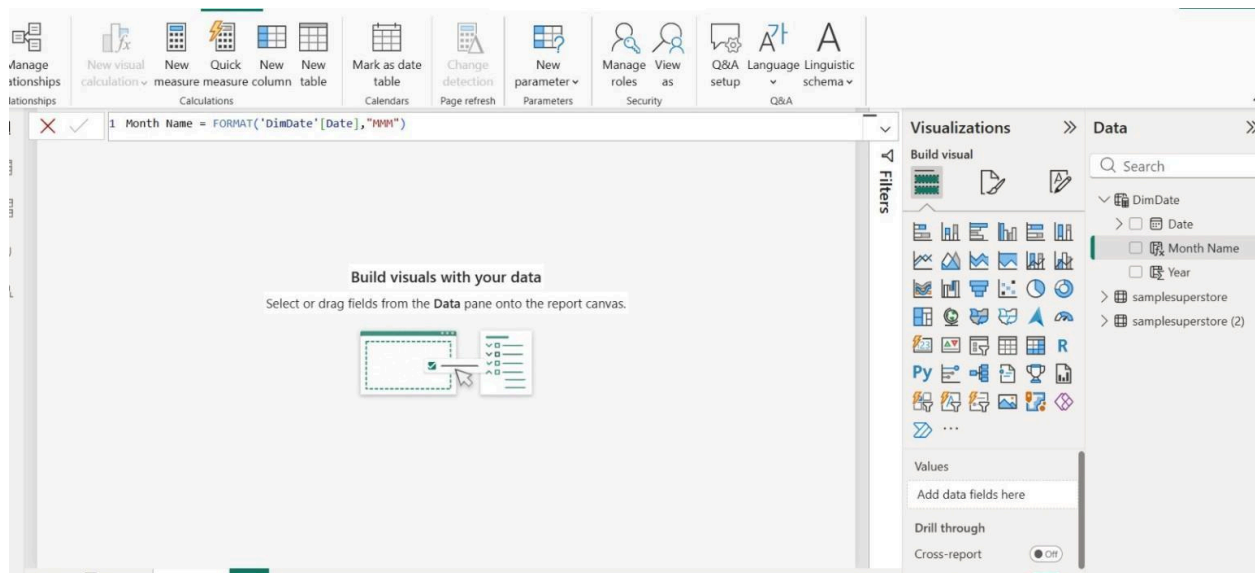


Figure 2: Creating the Month Name field in the DimDate table.

- Date: Main calendar date field.
- Year: Used for yearly analysis.
- Month Name: Displays the month in a readable three-letter format.
- These fields can be used for charts, slicers, and time-based analysis.

4. Data Model and Measures

The stock dataset and the supporting tables were organized in the Power BI model. A separate Measures Table was used to keep the analytical measures organized.

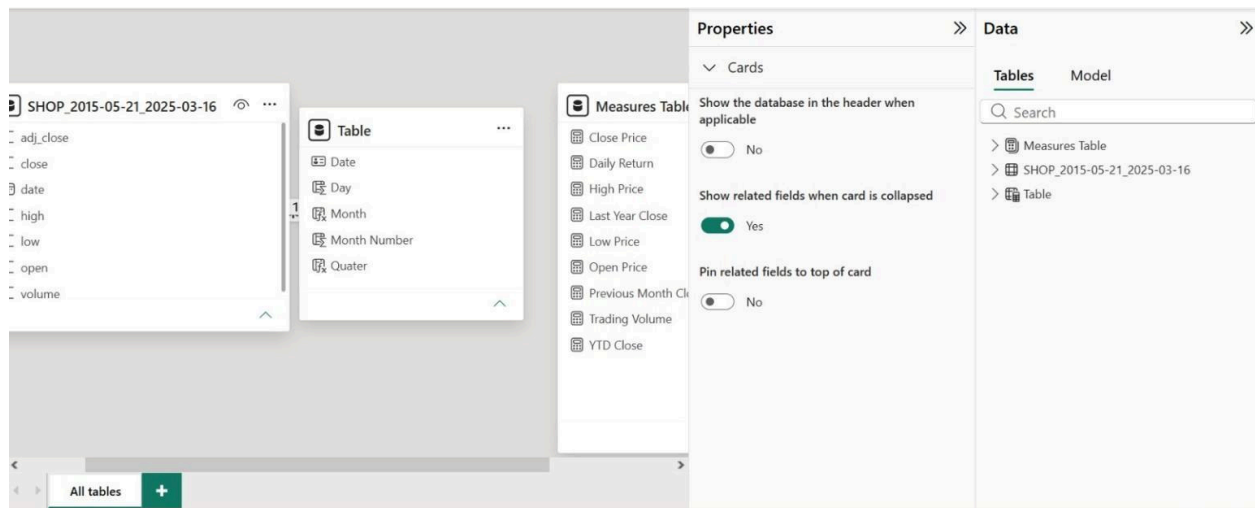


Figure 3: Power BI data model showing the stock data, date table, and measures table.

The project includes measures for key stock indicators. These measures were then used in card visuals and charts to summarize the data.

- High Price – displays the average high price.
- Low Price – displays the average low price.

- Close Price – displays the average closing price.
- Trading Volume – summarizes trading volume.
- Daily Return – calculates the daily change in closing price.
- YTD Close – supports year-to-date closing price analysis.

5. Stock Market Analysis Dashboard

The final report page was designed as a Stock Market Analysis Dashboard. Card visuals were used to display the main indicators, while line charts were used to show trends over time. A date slicer was also added so that the user can filter the dashboard by a selected date range.

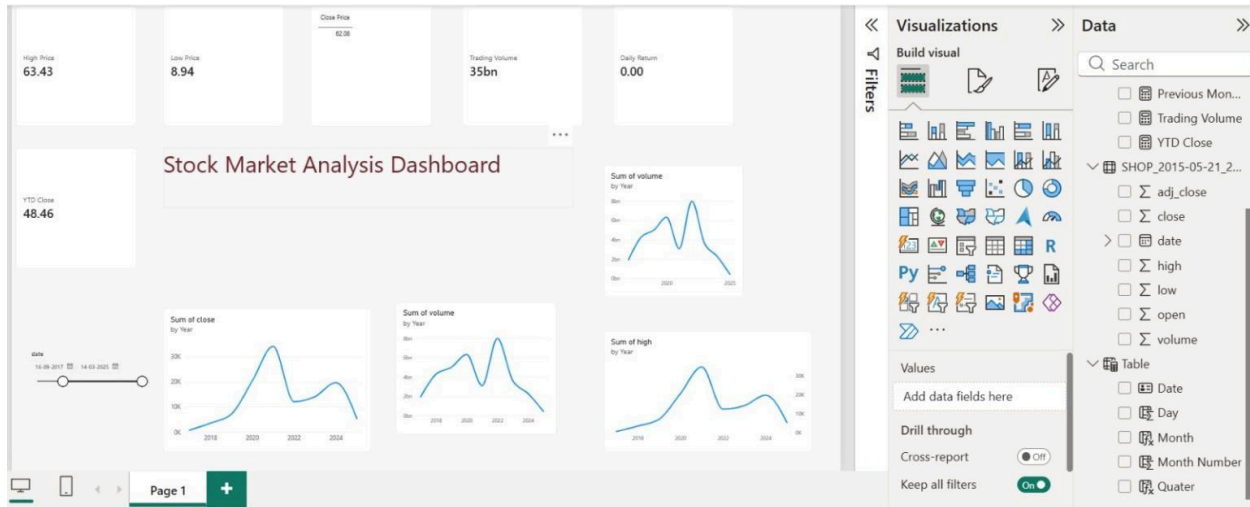


Figure 4: Final Stock Market Analysis Dashboard.

- High Price card shows the calculated high-price indicator.
- Low Price card shows the calculated low-price indicator.
- Close Price card shows the closing-price indicator.
- Trading Volume card summarizes the trading activity.
- Daily Return card displays the daily return measure.
- YTD Close card displays the year-to-date close measure.
- Line charts show stock price and volume trends.
- The date slicer allows interactive filtering of the report.

6. Power BI Visual and Card Layout Reference

The following screenshot shows an earlier Power BI report page containing card visuals and measures. It is included as a visual reference for the dashboard-building approach used during the project.

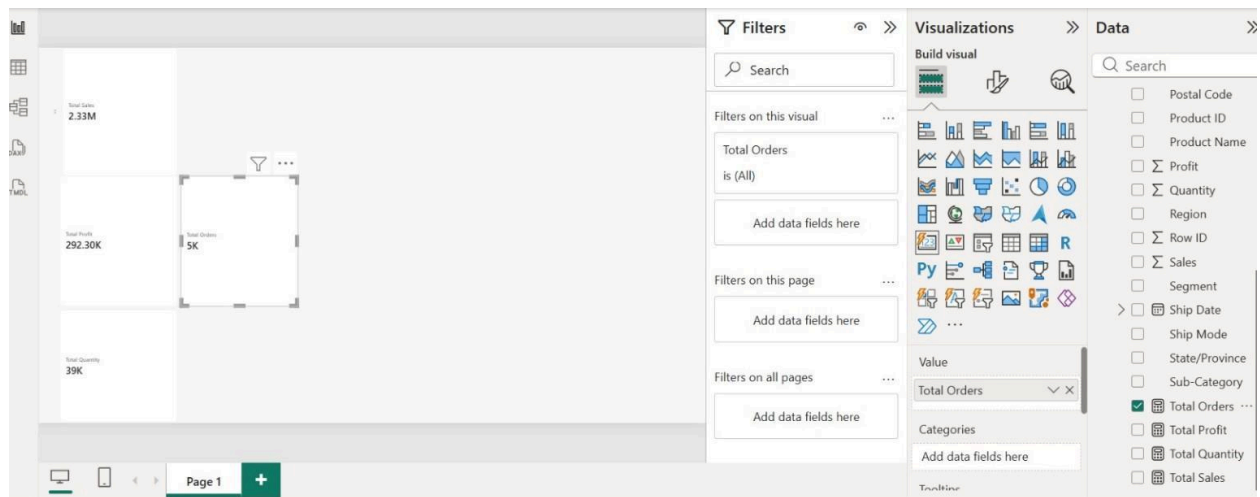


Figure 5: Power BI card visual and measure layout reference.

This reference demonstrates how KPI cards can be arranged on a report page to provide quick summaries before or alongside analytical charts.

7. Conclusion

The Week 3 Power BI project successfully transformed stock market data into an interactive analytical dashboard. The project covered data preparation, date-table creation, calculated measures, KPI cards, charts, and filtering. The final dashboard provides a clear view of stock prices and trading volume and demonstrates how Power BI can be used to convert raw financial data into meaningful visual insights.

End of Report